

Modified Photon Dispersion and Cosmic Gamma-Ray Delays in SDGFT

David A. Besemer^{1,*}

¹*Independent Researcher*

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We derive the modified photon dispersion relation from the discrete 24-cell lattice structure of spacetime. We calculate the cosmological propagation delays of high-energy photons, showing that the LIV parameter is $\xi_{LIV} = \sqrt{24/67} \approx 0.5985$. We compare our results to GRB 090510, showing that the predicted delay of 0.63 s complies with Fermi-LAT limits.

I. INTRODUCTION

Lorentz invariance violation (LIV) is a common prediction of quantum gravity. We derive the modified dispersion relation for photons in SDGFT.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$E^2 = p^2 c^2 \left(1 + \xi_{LIV} \frac{E}{E_{Pl}} \right) \quad (\text{modified dispersion relation}) \quad (1)$$

$$\Delta t = \xi_{LIV} \frac{E}{E_{Pl}} \int_0^z \frac{1+z'}{H(z')} dz' \approx 0.63 \text{ s} \quad (\text{for GRB 090510 at 31 GeV}) \quad (2)$$

III. GAMMA-RAY BURST DELAY

We integrate the propagation delay over cosmological distances under the Horndeski background, showing that the delay is within the Fermi-LAT upper limits.

IV. CONCLUSION

The predicted delay is consistent with Fermi-LAT and MAGIC observations, and will be tested by the Cherenkov Telescope Array (CTA).

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- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

* david.besemer@sdgft.org